Phase 2: Requirement Analysis

1. Introduction

The Requirement Analysis phase defines the functional, non-functional, and data requirements for the Rising Waters Flood Prediction System.

2. Functional Requirements

These define what the system *should do*: * **Data Input:** Users must be able to input the following features manually through a web form: Annual Rainfall, Seasonal Rainfall, Temperature, Humidity, and Cloud Visibility. * **Prediction Generation:** The system must process the input data using the pre-trained Machine Learning model. * **Result Display:** The application must display the binary outcome (Flood / No Flood) and the calculated probability percentage. * **User Management (Future/Optional):** The system should support basic user roles (Admin vs. Standard User) for tracking who is making predictions.

3. Non-Functional Requirements

These define *how the system should perform*: * **Accuracy:** The underlying predictive model must maintain an accuracy score of >90% (achieved ~96.55% with XGBoost). * **Usability:** The web application must be responsive and accessible on standard desktop and mobile browsers. * **Performance:** Predictions should be returned in under 2 seconds. * **Deployment:** The application structure must remain compatible with standard deployment platforms (e.g., Heroku, Render) without requiring root-level structural changes.

4. Data Requirements

- **Data Source:** Historical Rainfall Dataset (Kaggle).
- **Features Required:**
  - `Temperature` (Float)
  - `Humidity` (Float)
  - `Cloud Cover` / `Visibility` (Float)
  - `Annual Rainfall` (Float)
  - `Seasonal Rainfall` (Jun-Sep) (Float)
- **Target Variable:** `Flood` (Binary Classification: 1/0 or Yes/No).

5. Software & Hardware Requirements

- **Programming Language:** Python 3.x
- **Libraries:** Pandas, NumPy, Scikit-learn, XGBoost, Flask
- **Frontend:** HTML5, CSS3
- **Development Environment:** Jupyter Notebooks (for EDA and modeling), VS Code / IDE (for application development).